

Building a Climate-Resilient Water Future for Bengaluru: A CRIDA Approach to Urban Water Scarcity and Flood Risk

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Introduction

1. Background and Problem Statement
2. Project Objective
3. Applying the CRIDA Five-Step Planning Process
4. CRIDA Stress Test
5. Most Challenging CRIDA Steps in Bengaluru
6. Conclusion

Introduction

As a Bangalorean, the water crisis our city has faced in recent years has made one thing clear: climate change is no longer a distant possibility—it is shaping our lives and our decisions today.

To better understand how communities and policymakers can navigate these uncertainties, I recently completed UNESCO's Introduction to Climate Risk Informed Decision Analysis (CRIDA) course. The course challenged me to think beyond traditional planning and embrace adaptive, risk-informed approaches to water and climate governance.

One of the most exciting parts of this journey is developing my own CRIDA project, applying these ideas to real-world challenges. As someone deeply interested in public policy and sustainable development, this learning experience has reinforced my belief that building climate resilience requires not only infrastructure and technology but also better decision-making frameworks.

The future of our cities depends on how well we prepare for uncertainty today. I look forward to continuing this journey and contributing to conversations on climate adaptation and water security.

Background and Problem Statement

Bengaluru, often called India's "Silicon Valley," is facing a severe urban water crisis. The city has experienced:

Depletion of groundwater (water tables declining by 20-30 meters in several areas).

Loss of nearly 85% of its historical lakes.

Increasing dependence on the distant Cauvery River.

Severe urban flooding due to unplanned urbanization.

Growing water demand from a population expected to exceed 16 million by 2035.

Increasing climate variability with irregular monsoons, droughts, and extreme rainfall events.

The 2024 Bengaluru water crisis, where many areas experienced acute shortages and tanker prices increased drastically, demonstrated the urgent need for climate-informed water planning.

Traditional water planning assumes a stable climate. However, climate uncertainty makes long-term infrastructure decisions difficult. Therefore, a CRIDA-based approach is highly suitable for Bengaluru.

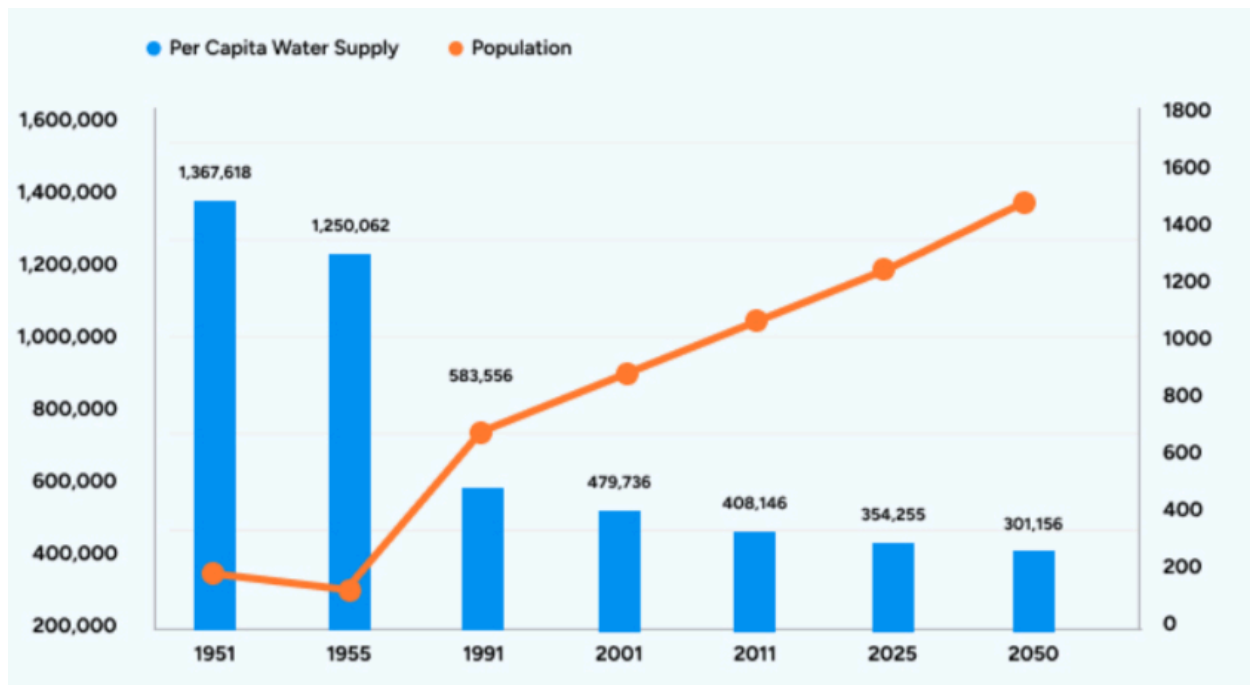
According to India Today, there is no shortage of water in areas serviced by Cauvery Water. Those areas dependent on borewells have been facing a crisis as borewells have dried up.

Of the nearly 13,900 borewells in Bengaluru, nearly 7,000 have dried up.

Despite receiving over 900 mm of rainfall annually, the city struggles to store an adequate amount of water.

Notably, Bengaluru is regarded as one of the most water-stressed cities globally, comparable to Cape Town, South Africa, which nearly faced a day zero scenario.

A BBC report, based on UN projections, listed Bengaluru as second after Brazil's São Paulo among the 11 global cities that are likely to run out of drinking water.



Project Objective

To develop a climate-resilient urban water management strategy that ensures:

Reliable water supply.

Reduced flood vulnerability.

Sustainable groundwater management.

Protection of urban ecosystems.

Flexible adaptation pathways under uncertain climate futures.

Applying the CRIDA Five-Step Planning Process

Step 1: Understand the Context

- Stakeholders
- Bangalore Water Supply and Sewerage Board (BWSSB)
- Bruhat Bengaluru Mahanagara Palike (BBMP)
- Karnataka State Disaster Management Authority
- Residents Welfare Associations
- Industries and IT companies
- Urban planners and environmental NGOs
- Researchers and universities

Key Questions

How much water will Bengaluru need by 2050?

How will climate change affect rainfall and drought?

What are the major vulnerabilities?

System Mapping

Drivers

Climate change + population growth + urbanisation

↓

Impacts

Water shortages, groundwater depletion, flooding

↓

Consequences

Economic losses, health risks, reduced quality of life.

Step 2: Assess Vulnerability and Climate Risks

Major Climate Risks

Hazard.	Impact
Drought.	Water shortages

Intense rainfall.

Urban flooding

Heat waves.

Increased water demand

Groundwater decline.

Supply insecurity

Climate Scenarios

- Scenario A

Moderate warming (+1.5°C)

- Scenario B

High warming (+2.5°C)

- Scenario C

Extreme climate variability.

Step 3: Identify Adaptation Options

- Structural Measures

Lake rejuvenation.

Managed aquifer recharge.

Wastewater recycling plants.

Decentralized sewage treatment systems.

Stormwater harvesting infrastructure.

Smart metering.

- Nature-Based Solutions

Wetland restoration.

Urban forests.

Green roofs.

Permeable pavements.

- Governance Measures

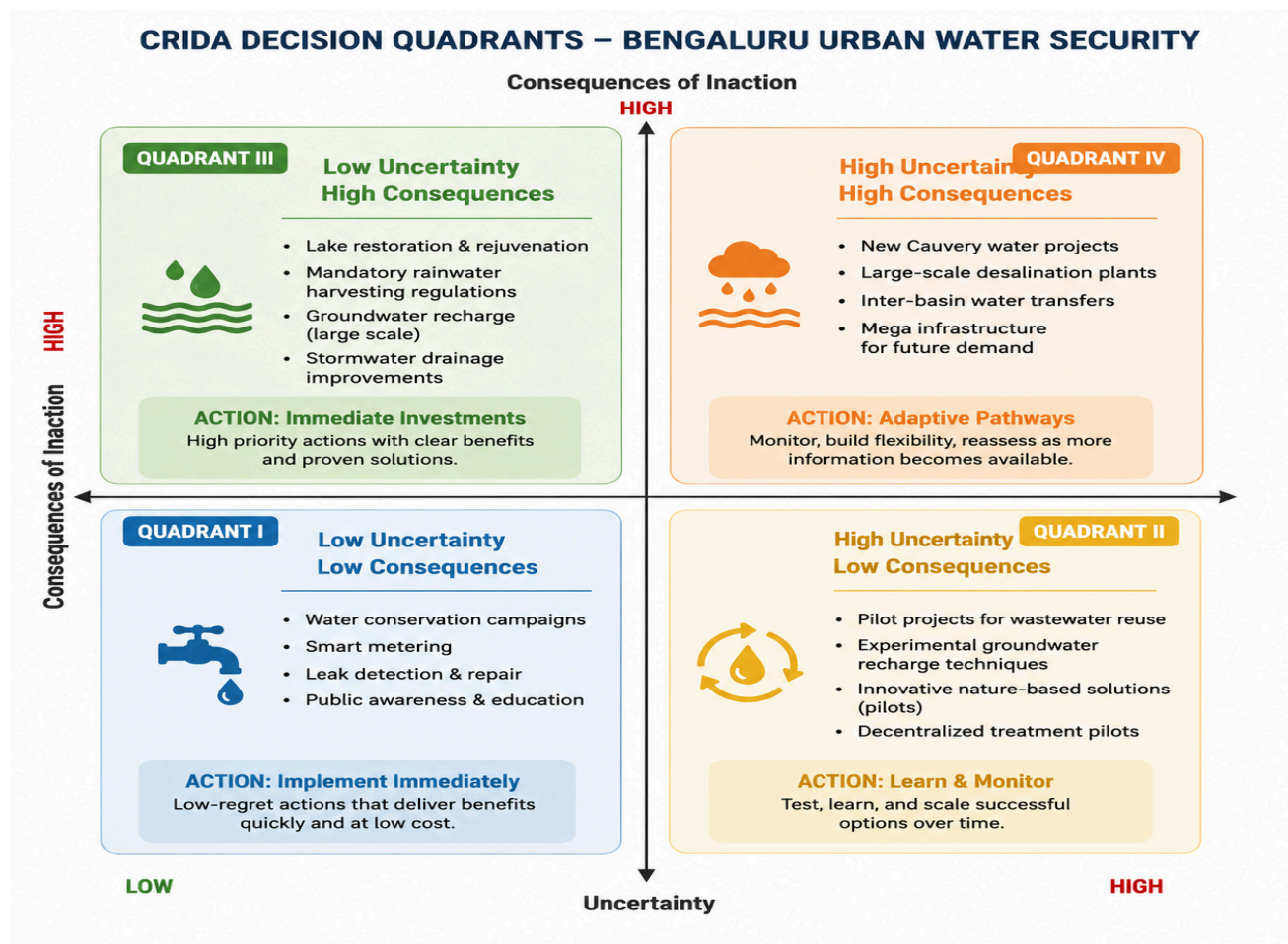
Water pricing reforms.

Mandatory rainwater harvesting.

Demand management.

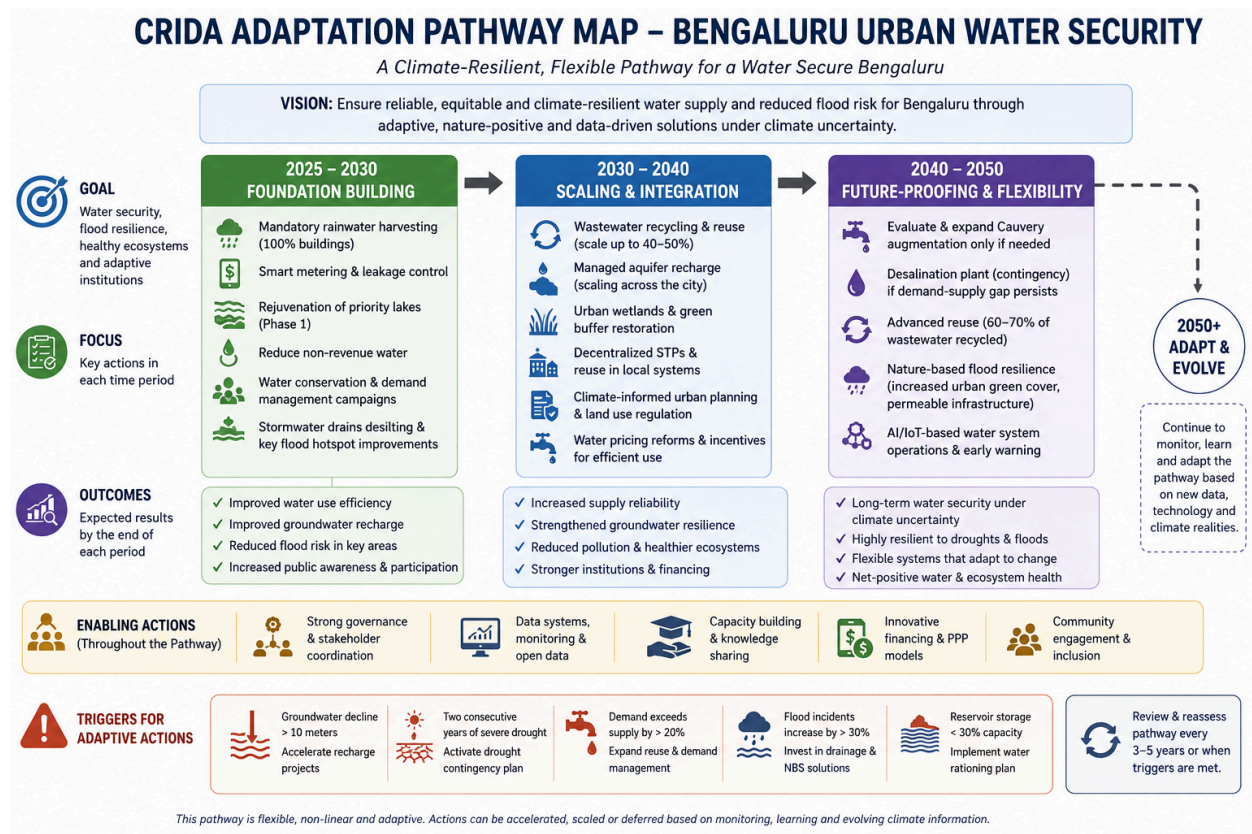
Climate-informed urban planning.

Step 4: Evaluate Adaptation Pathways



Adaptation pathway map

The pathway approach avoids maladaptation and lock-in by allowing flexible decisions as climate information improves.



Step 5: Implement, Monitor and Adjust

Indicators

Groundwater levels.

Reservoir storage.

Percentage of recycled water.

Flood incidents.

Per capita water availability.

Number of rejuvenated lakes.

Trigger Points

Indicator.

Threshold

Groundwater decline >10m.

Accelerate recharge projects

Demand exceeds supply by 20%.

Expand wastewater reuse

Severe drought for 2 year.

Activate contingency plans

Flood incidents increase by 30%.

Invest in drainage upgrades

CRIDA Stress Test

The system is tested under:

Extreme drought.

Consecutive monsoon failures.

Population growth above projections.

Extreme rainfall events.

The objective is to identify adaptation tipping points and revise strategies before system failure occurs.

Most Challenging CRIDA Steps in Bengaluru

1. Stakeholder Coordination

Multiple agencies work independently, creating governance fragmentation.

2. Climate Data Availability

High-resolution climate projections for Bengaluru are limited.

3. Financing

Large investments are required for resilient infrastructure.

4. Political and Institutional Continuity

Long-term adaptation measures often change with political priorities.

Areas Where More Support is Needed

I would require additional support in:

Hydrological modelling.

Downscaled climate projections.

Economic cost-benefit analysis.

Designing adaptation pathways and tipping points.

GIS and scenario modelling.

Expected Outcomes

By applying CRIDA, Bengaluru can:

- ✓ Improve water security.
- ✓ Reduce flood risks.
- ✓ Increase groundwater resilience.
- ✓ Strengthen institutional coordination.
- ✓ Build a flexible, climate-resilient urban water system.

Conclusion

The Bengaluru water crisis demonstrates that cities can no longer rely on traditional planning methods based on historical climate patterns. A CRIDA approach, combining uncertainty management, adaptation pathways, and robust decision-making, provides a practical framework for creating a resilient urban water future. Bengaluru can become a global example of climate-resilient urban water governance, showing how rapidly growing cities can adapt to increasing climate uncertainty while ensuring sustainable and equitable water access for all citizens.